Ingeniería de Software I

Introducción a Containers, Docker, Kubernetes (K8s)

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Agenda - Containers, Docker y K8s

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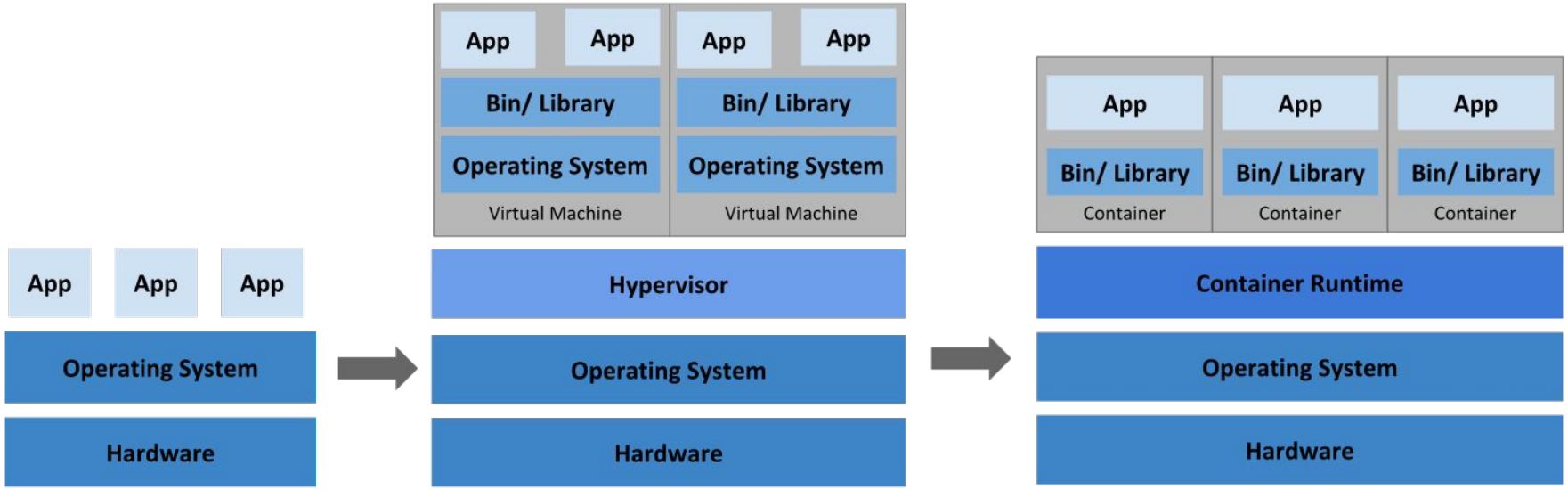
1. Introducción
2. VMs vs Containers
3. Docker
4. Arquitectura y Automatización
5. Kubernetes
6. Cloud Native Platforms
7. Links de interés

VMs vs Containers

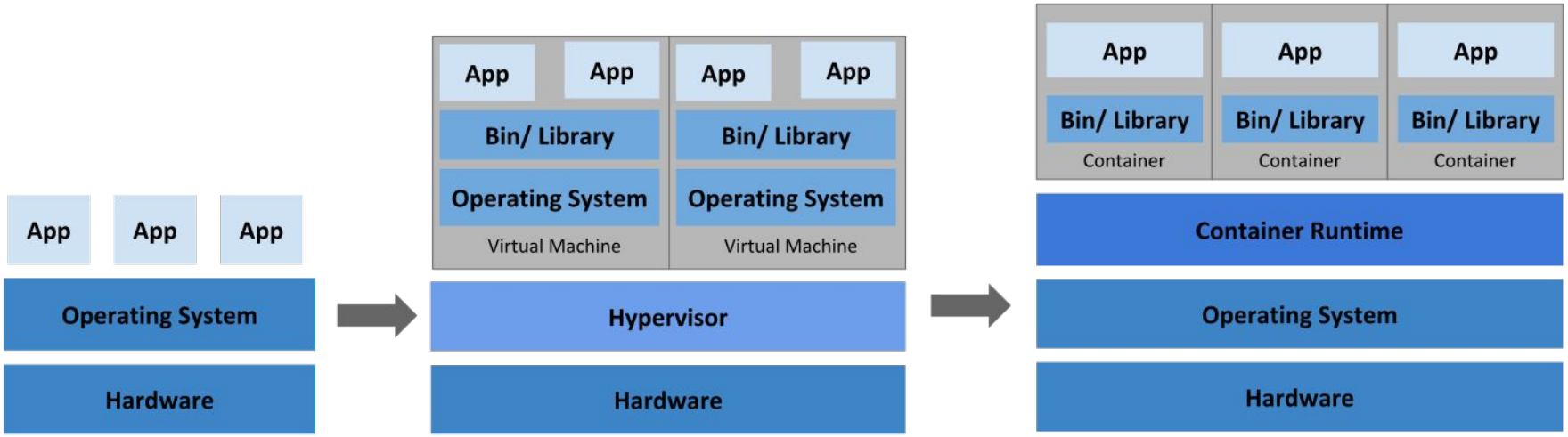


Traditional Deployment

Virtualized Deployment
Hosted Hypervisor

Container Deployment

VMs vs Containers



Traditional Deployment

Virtualized Deployment
Bare-Metal Hypervisor

Container Deployment

VMs Players

vmware
vSphere
Microsoft
Hyper-V
redhat
solaris
CITRIX
XenServer
AIX
ORACLE
VM
NUTANIX
KVM

Docker

Docker Beneficios

- Portabilidad y multi cloud
- ROI
- Estandarización y Productividad
- Rápido desarrollo y mayor eficiencia con CI/CD
- Compatibilidad y mantenibilidad.
- Simplicidad
- Testing automatizado
- Mayor calidad de productos

Docker in numbers

CONTAINERS ARE NOW MAINSTREAM AND USAGE IS ONLY GROWING.

242B
Total Pulls on Hub

11B
Pulls in the past month*

7M
Repositories on Hub

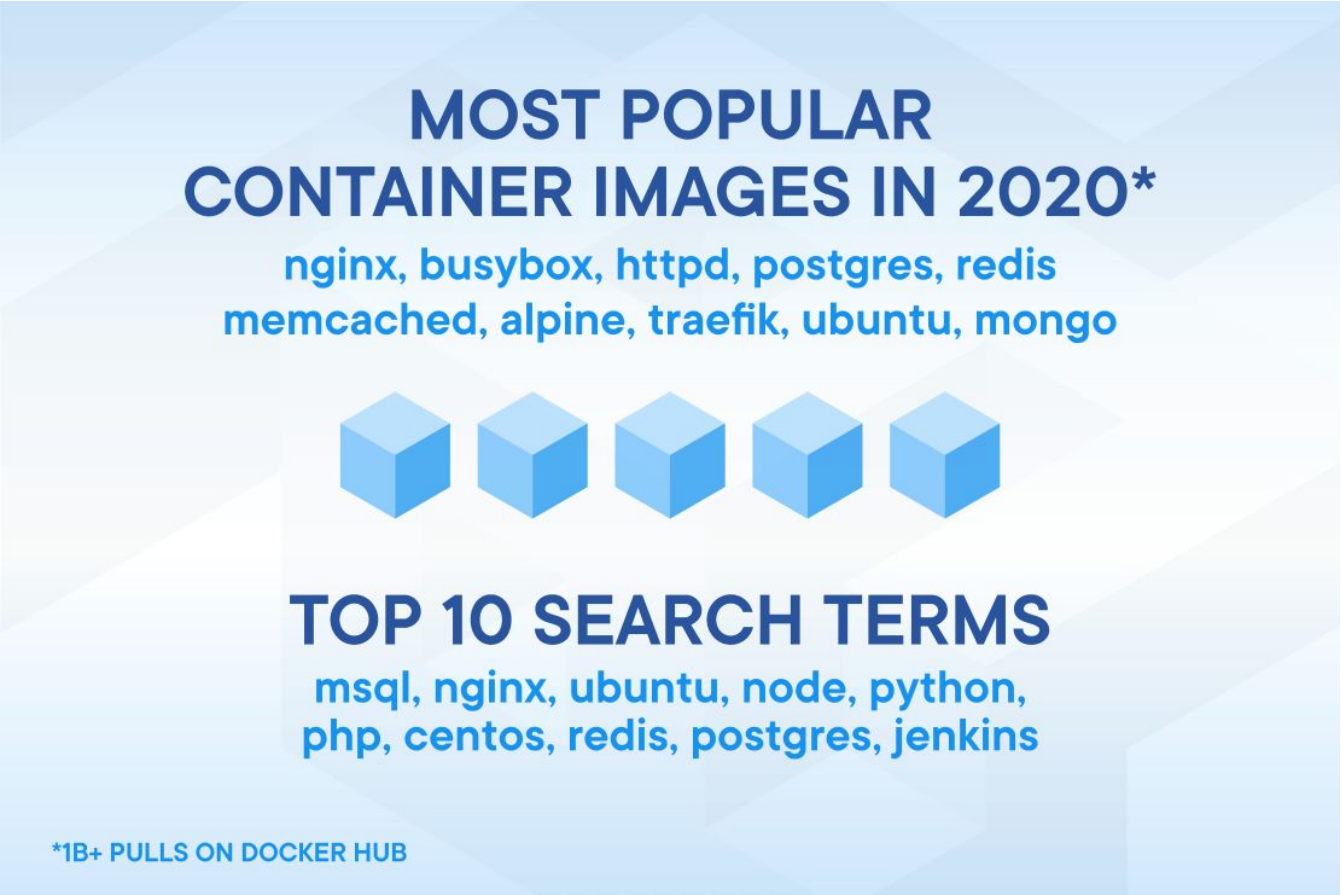
7M
Hub Users

2.9M
Desktop Installations

*UP FROM 5.5B A YEAR AGO

Ver: https://www.docker.com/blog/introducing-the-docker-index/

Docker in numbers



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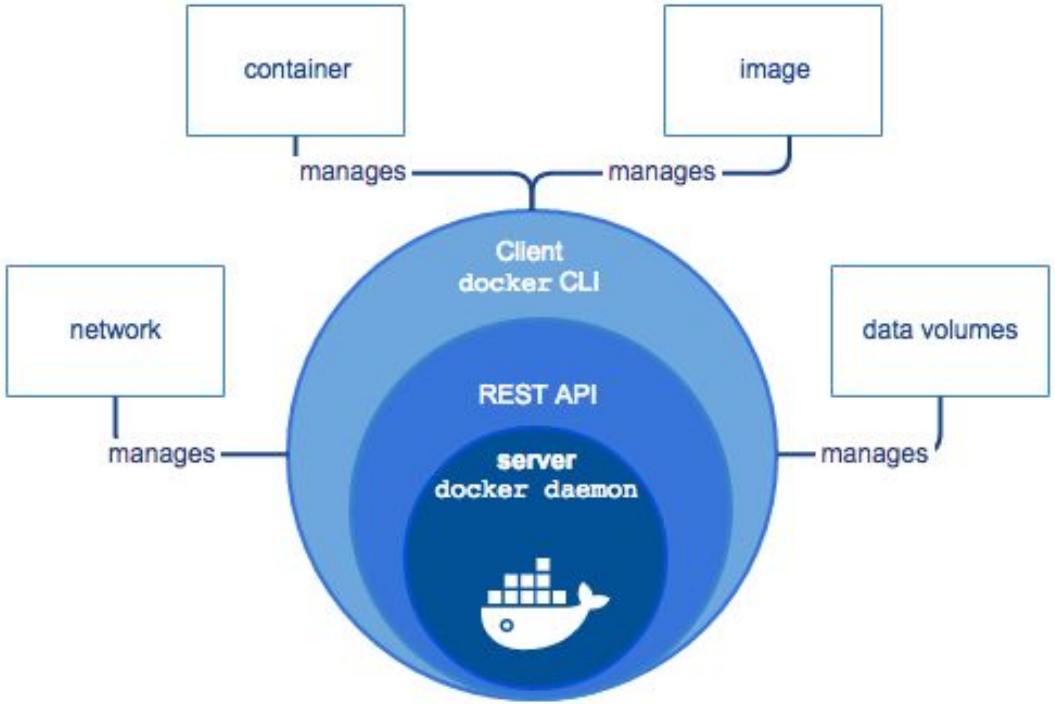
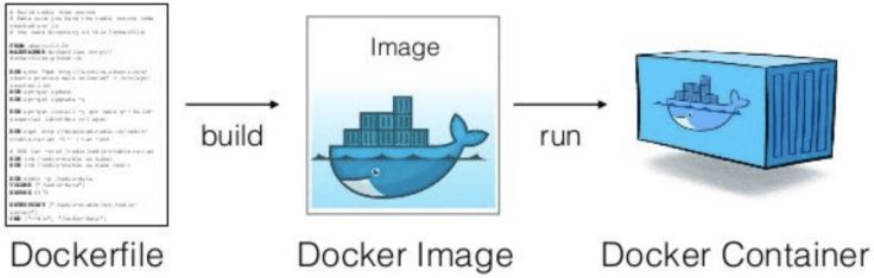
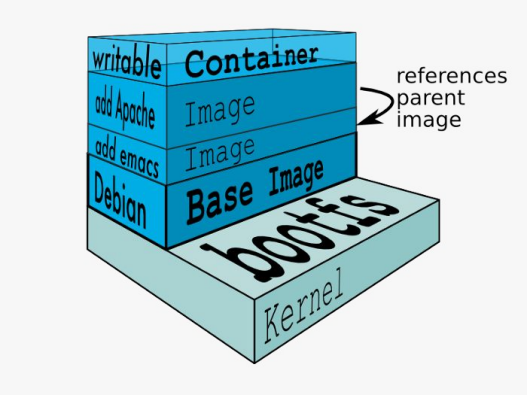
Docker Architecture

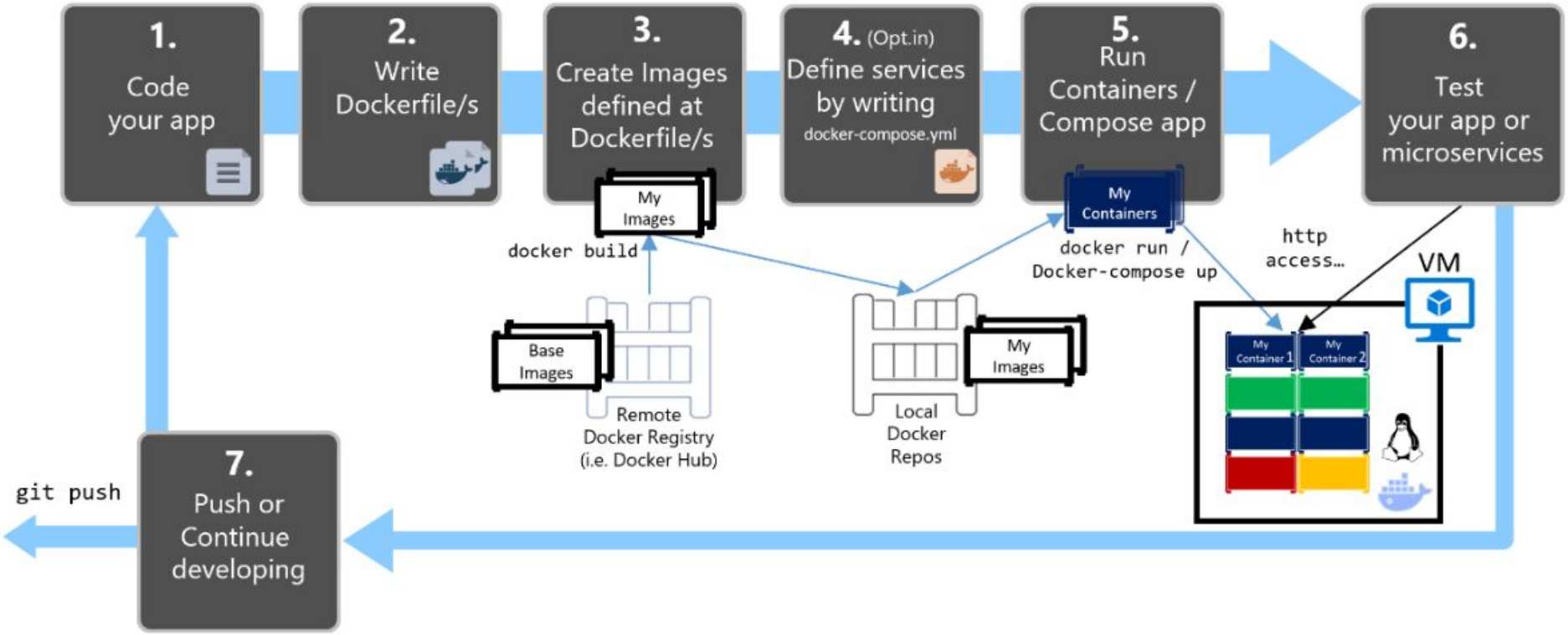


Docker Architecture





Docker developing workflow



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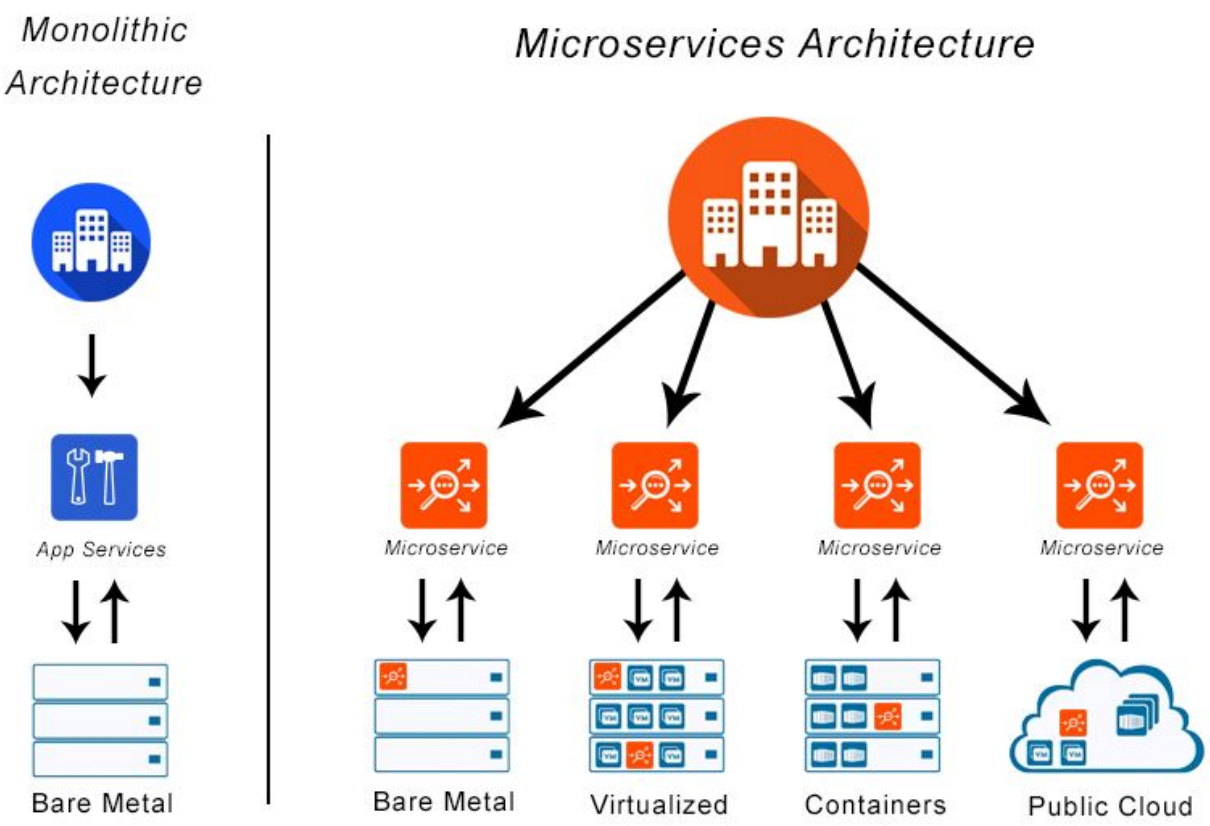
Arquitectura y automatización

Service Oriented Architecture



Methodology: Docker EE Modernizes Apps and Infrastructure

The quickest way to cut into that 80%

Existing Application

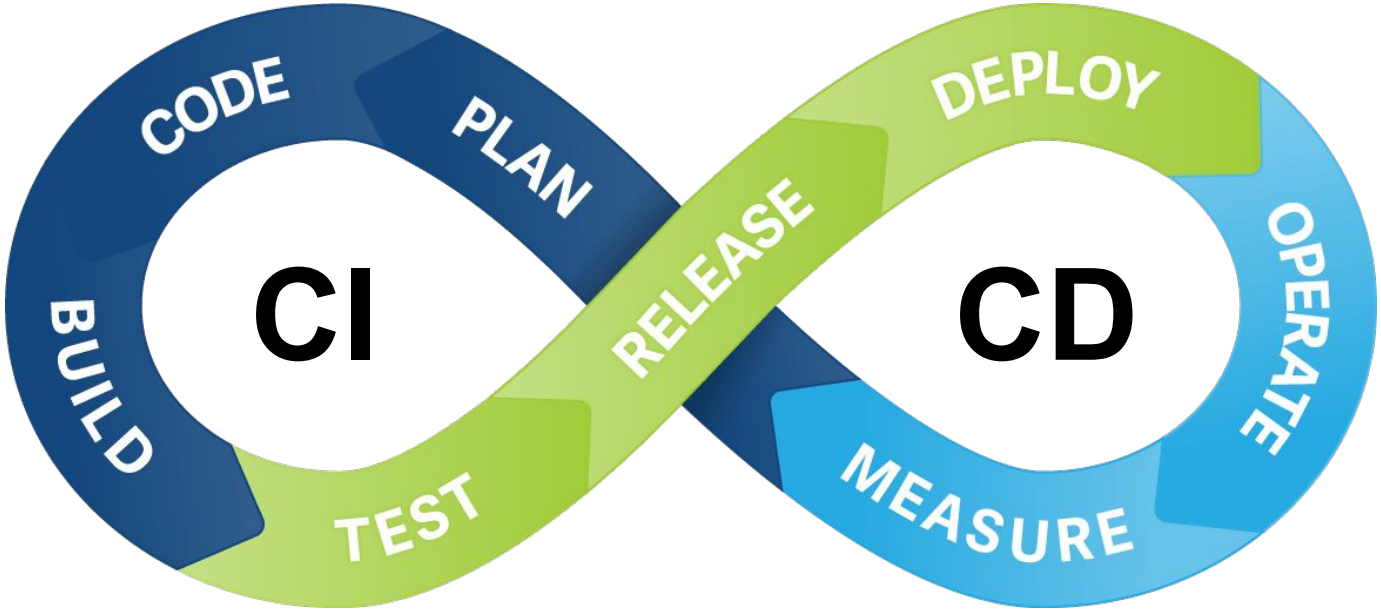
Convert to a **container** with Docker EE

Modern Infrastructure
Built on premise, in the cloud, or as part of a hybrid environment.

Modern Methodologies
Integrate to CI/CD and automation system

Modern Microservices
Add new services or start peeling off services from monolith code base

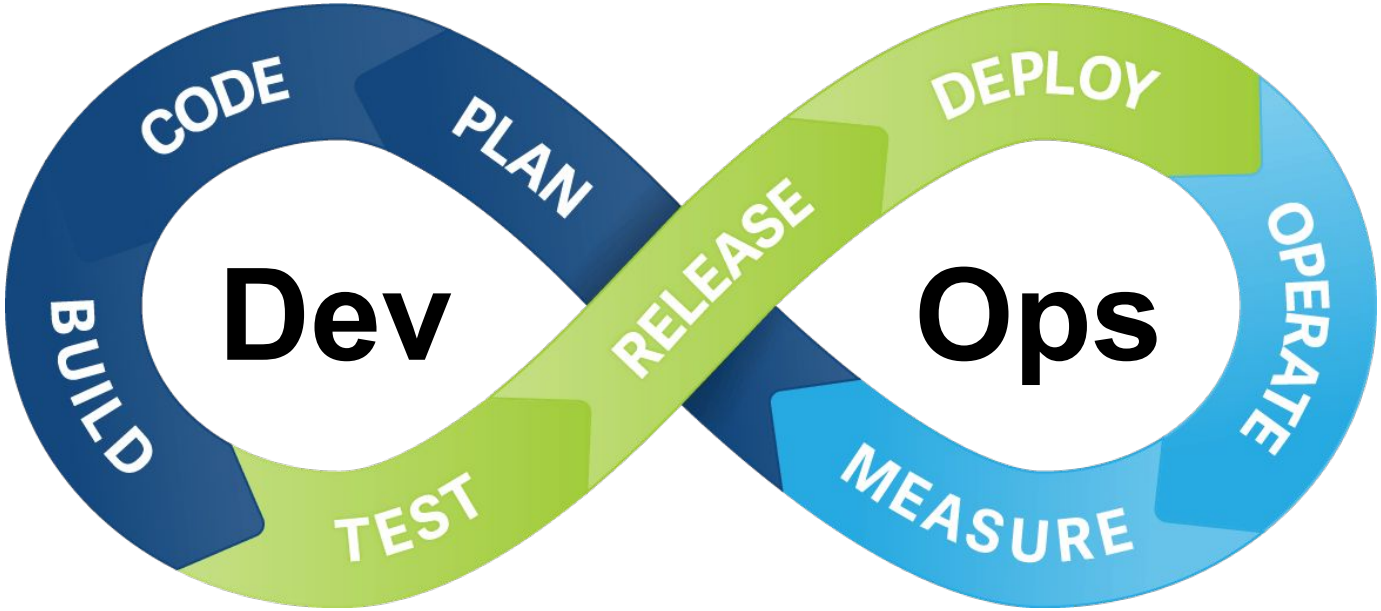
Faster Delivery



Continuous Integration

Continuous Delivery

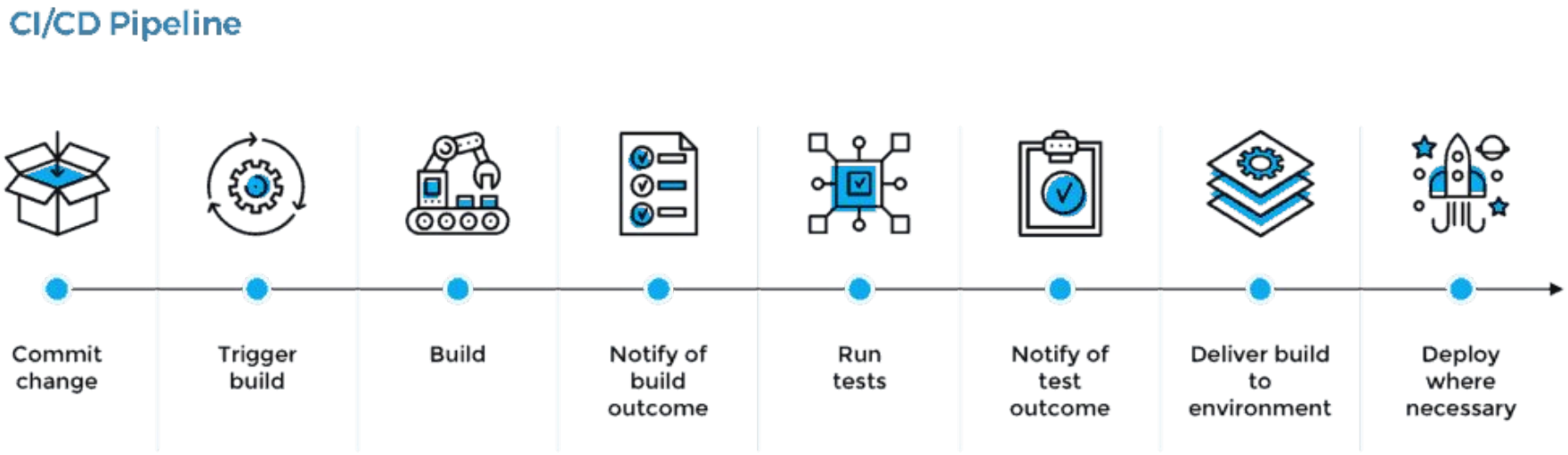
Faster Delivery



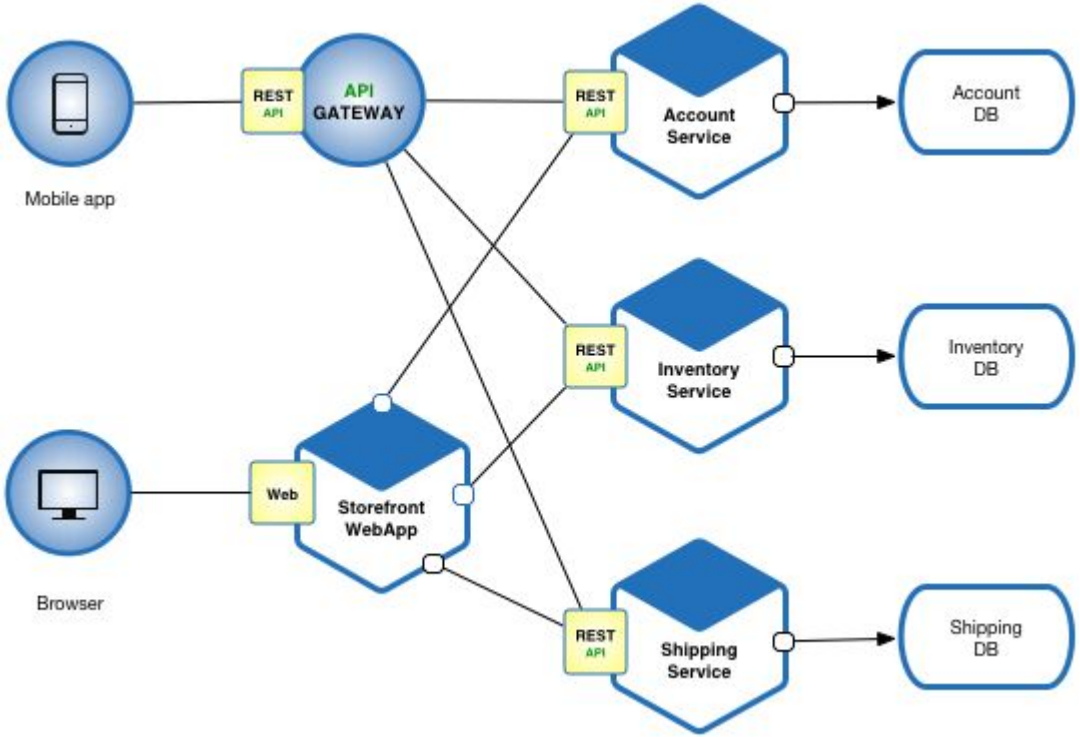
Continuous Integration

Continuous Delivery

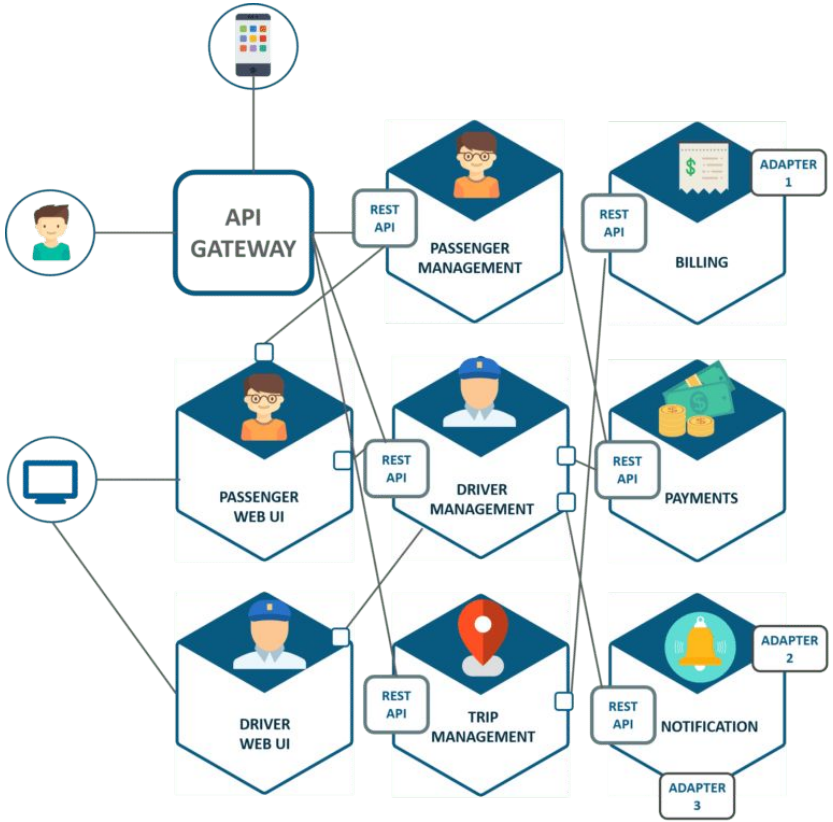
Faster Delivery pipelines



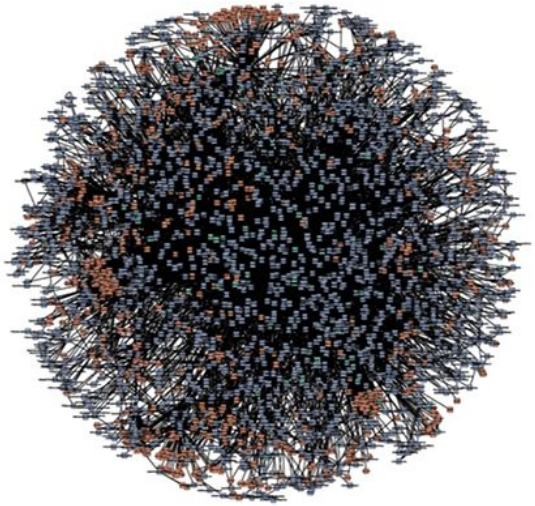
Arquitectura de Microservicios - Base



Arquitectura de Microservicios - Uber



Arquitectura de Microservicios - Netflix y Amazon



amazon.com

NETFLIX

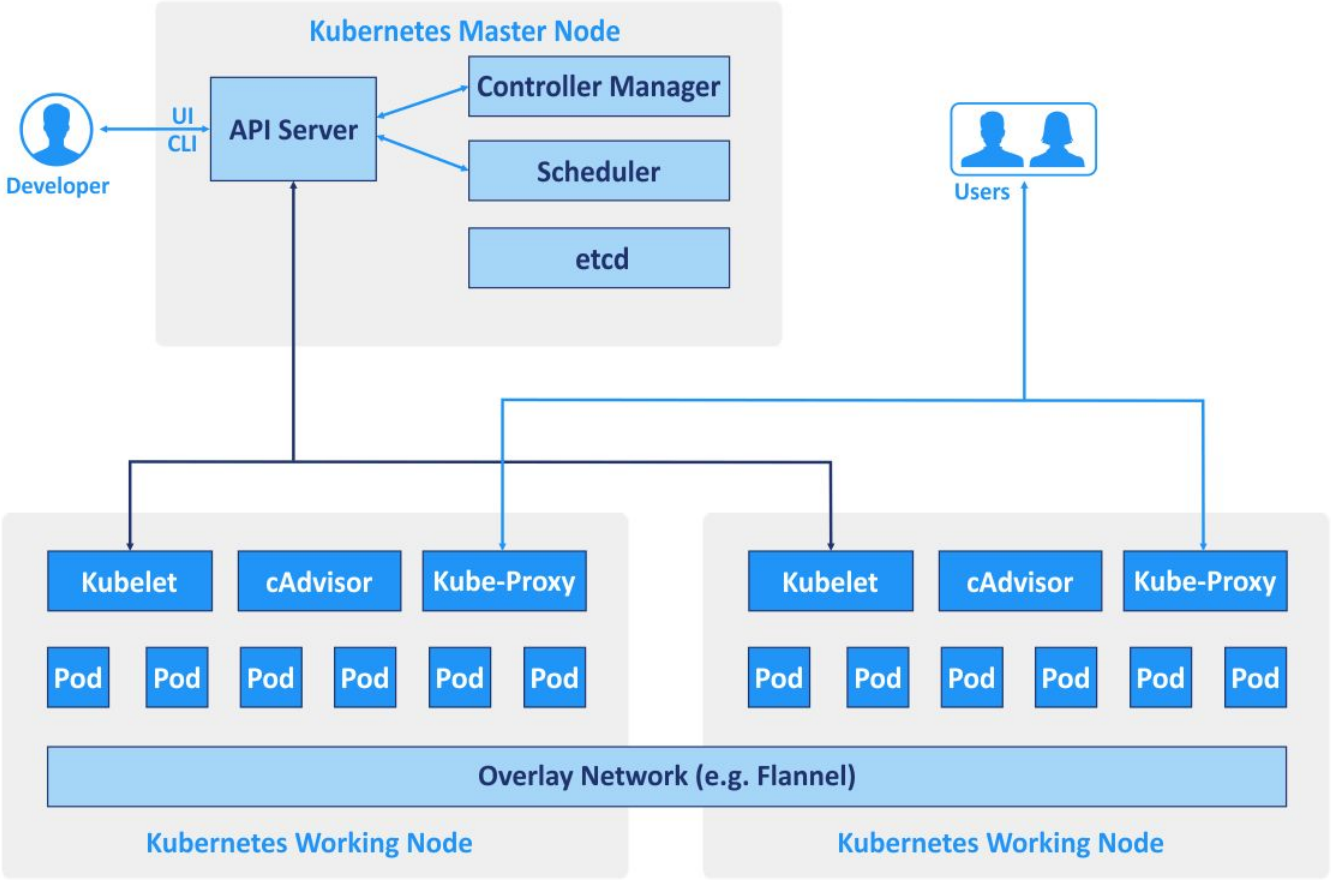
Kubernets (K8s)

Kubernetes - K8s - Qué es?

- Kubernetes is a portable, extensible, open-source platform for managing containerized workloads and services, that facilitates both declarative configuration and automation. It has a large, rapidly growing ecosystem. Kubernetes services, support, and tools are widely available.
- The name Kubernetes originates from Greek, meaning helmsman or pilot. Google open-sourced the Kubernetes project in 2014. Kubernetes combines over 15 years of Google's experience running production workloads at scale with best-of-breed ideas and practices from the community.
- It works for private, public or hybrid clouds.

Kubernetes - Benefits

Kubernetes - Architecture



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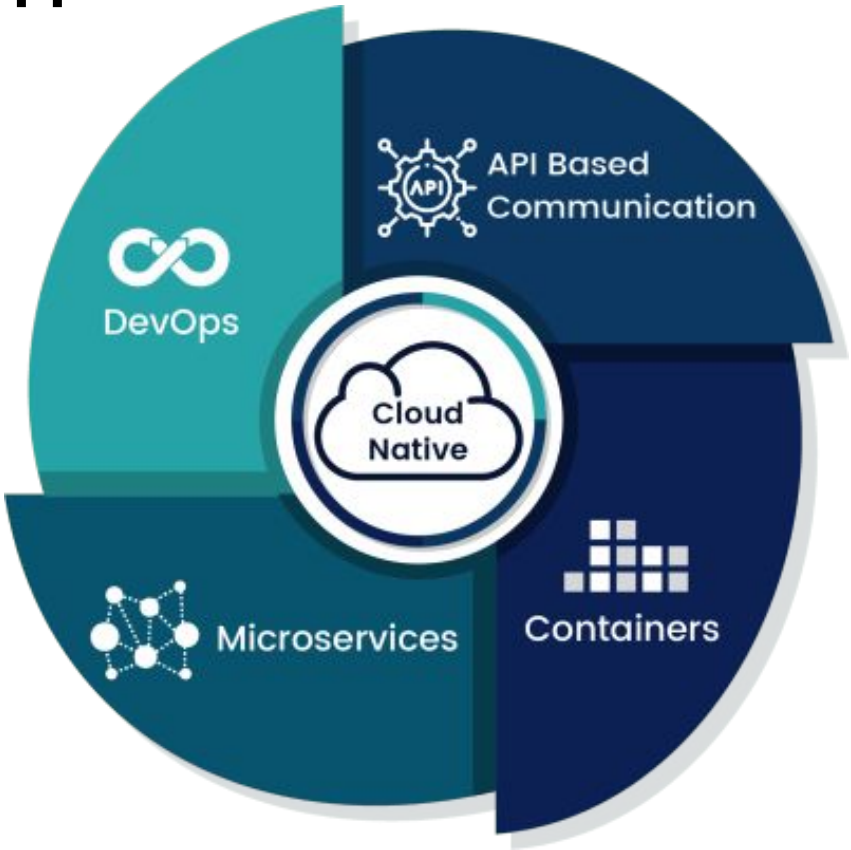
Cloud Native Platforms

Cloud Native Apps



Ver: https://landscape.cncf.io/

Cloud Native Maturity Model

Cloud Native

- Microservices architecture
- API-first design

Cloud Resilient

- Fault-tolerant and resilient design
- Cloud-agnostic runtime implementation
- Bundled metrics and monitoring
- Proactive failure testing

Cloud Friendly

- 12 Factor App methodology
- Horizontally scalable
- Leverages platform for high availability

Cloud Ready

- No permanent disk access
- Self-contained application
- Platform-managed ports and networking
- Consumes platform-managed backing services

Links de interés

Links de interés - Containers, Docker, K8s

- Docker
- Docker Hub
- Kubernetes (K8s)
- Cloud Native
- Cloud Native Landscape